Capstone Project 1 Report

" Uber Trip Analysis Dashboard"

Domain: Uber Trip Analytics

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1. Introduction :

In today's fast-growing ride-sharing industry, companies like Uber must efficiently manage trip demand, driver allocation, and revenue generation. Understanding travel patterns, peak demand hours, cancellation behaviour, and high-traffic zones is crucial for operational success.This project focuses on analyzing Uber trip data and building an interactive Power BI dashboard to extract meaningful insights that support data-driven decision-making.

2. Problem Statement :

Ride-hailing businesses like Uber operate in a highly dynamic environment where demand, pricing, and operational efficiency fluctuate based on time, location, and customer preferences. Without a centralized analytical solution, it becomes difficult to track booking

trends, optimize pricing strategies, and ensure efficient driver allocation. This project addresses the need for a comprehensive BI solution that enables stakeholders to monitor trip performance, identify demand patterns, and improve customer satisfaction through actionable insights.

3. Objective:

The primary goal of this project is to:

• Analyse historical Trip data to extract meaningful insights.
• Visualize KPIs to monitor Trip performance.
• To identify peak booking hours, high-demand locations, and supply–demand gaps in order to optimize driver allocation and reduce wait times.

4. Tools & Technologies:

Tool	Purpose
MySQL	Data transformation and SQL querying for cross verifying
Python (Jupyter Notebook)	SQL result validation via Python-MySQL connector, Data cleaning,tranformation
Power BI	Visual dashboard creation and interactive reporting
CSV File	Source data used for analysis

5. Methodology :

1. Data Import & Cleaning:

o Imported CSV data into MySQL using proper schema.
o Aligned data types for accurate analysis.

2. Data Analysis (SQL):

o Wrote SQL queries to calculate KPIs such as total applications, Total bookings, Average booking value,Total trip distance,Average trip distance,Average trip Time etc.

3. Validation in Python:

o Connected MySQL to Python using pymysql.

o Executed queries and cross-verified results from SQL against Power BI measures.

4. Dashboard Development (Power BI):

o Imported cleaned data.
o Built visualizations including bar charts, stacked charts, slicers, Heatmaps, and KPI cards.
o Added filters for Locations, Date, Payment_type, City.

6. Data Description :

Source: uber_trip_data.csv
Total Rows: 1,03,728

Key Fields: *Uber Trip Dataset*

• **trip_id –** Unique identifier for each trip.

• **pickup_datetime –** Date and time when the trip started (used for peak hour and trend analysis).

• **dropoff_datetime –** Date and time when the trip ended (used to calculate trip duration).

• **trip_status –** Indicates whether the trip was Completed, Cancelled, or Incomplete.

• **trip_distance –** Total distance covered during the trip.

• **fare_amount –** Total fare charged for the trip (revenue analysis).

• **payment_type –** Mode of payment (Cash, Card, Online, etc.).

• **driver_id –** Unique identifier for the driver (driver performance analysis).

• **pickup_location_id –** Location ID where the trip started.

• **dropoff_location_id –** Location ID where the trip ended.

Key Fields: *Location Dataset*

• **Location_id –** Unique identifier for each location zone.

• **Location -** Specific pickup/drop-off zone.

• **City -** City division where the trip occurred.

7. Dashboard Design Sections:

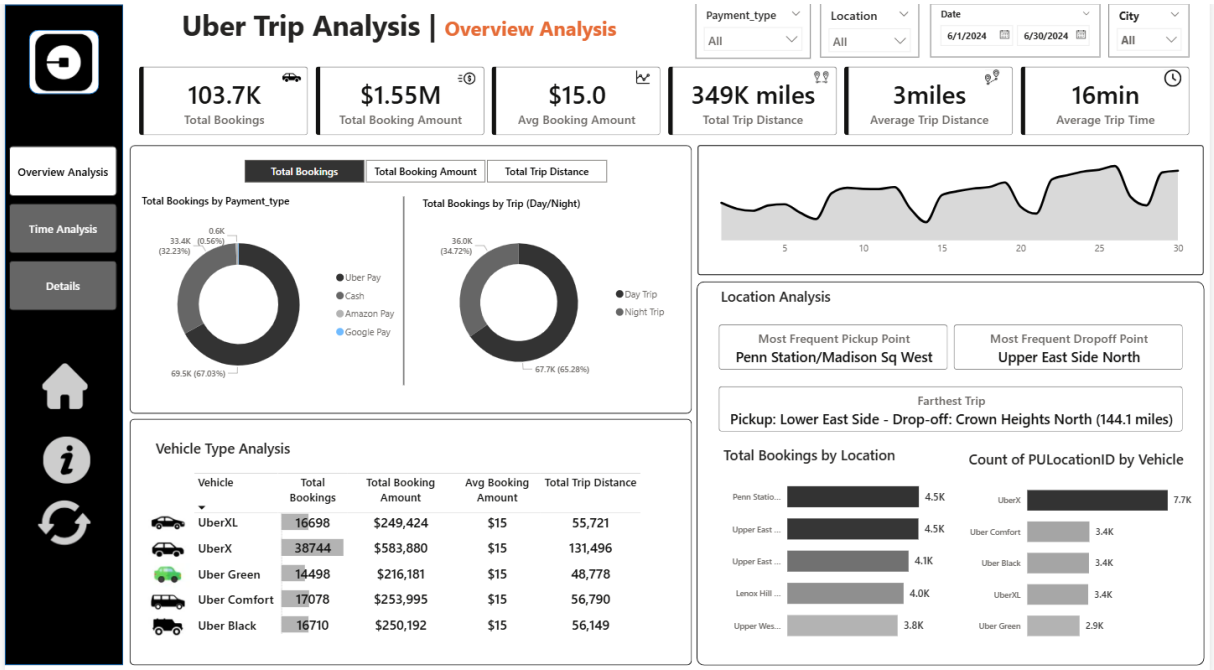
1. KPI Cards: Total Bookings, Total Booking Amount, Average Booking Amount, Total Trip Distance, Avgerage Trip Distance, Average Trip Time

2. Time Based Analysis: Trips by Hour, Trips by Time Slot, Monthly Trend Analysis.

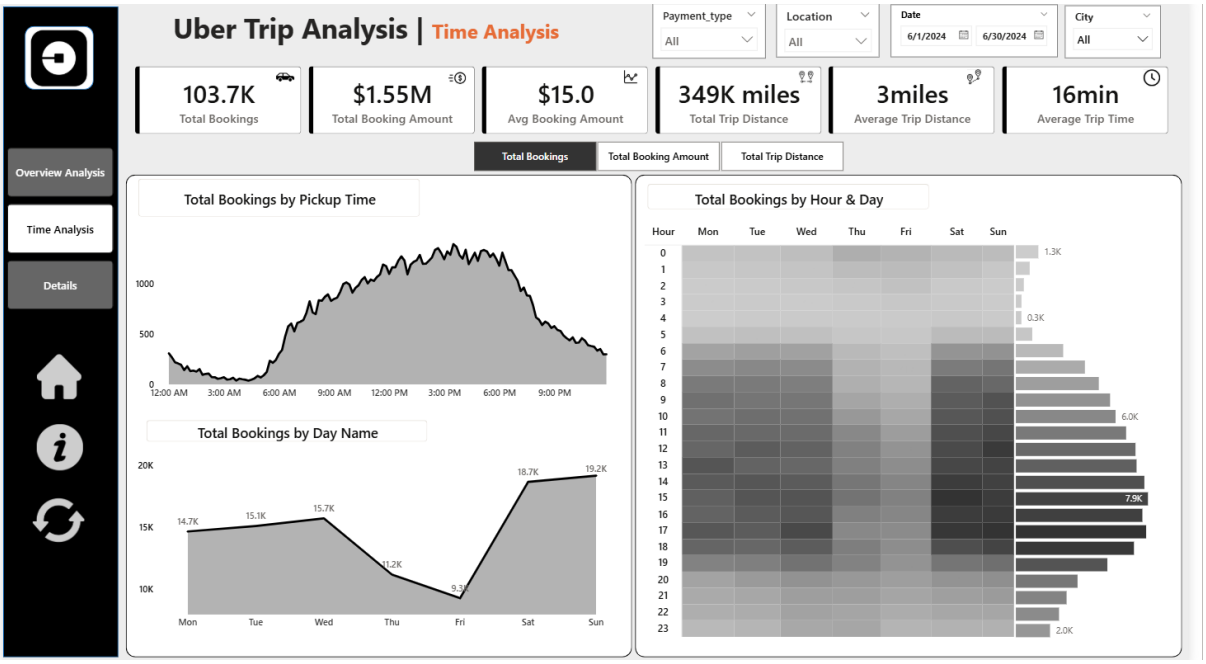
3. Location Analysis Bar charts: Pickup & Drop-off.

4. Filters: Date, Location, City, Payment_Type.

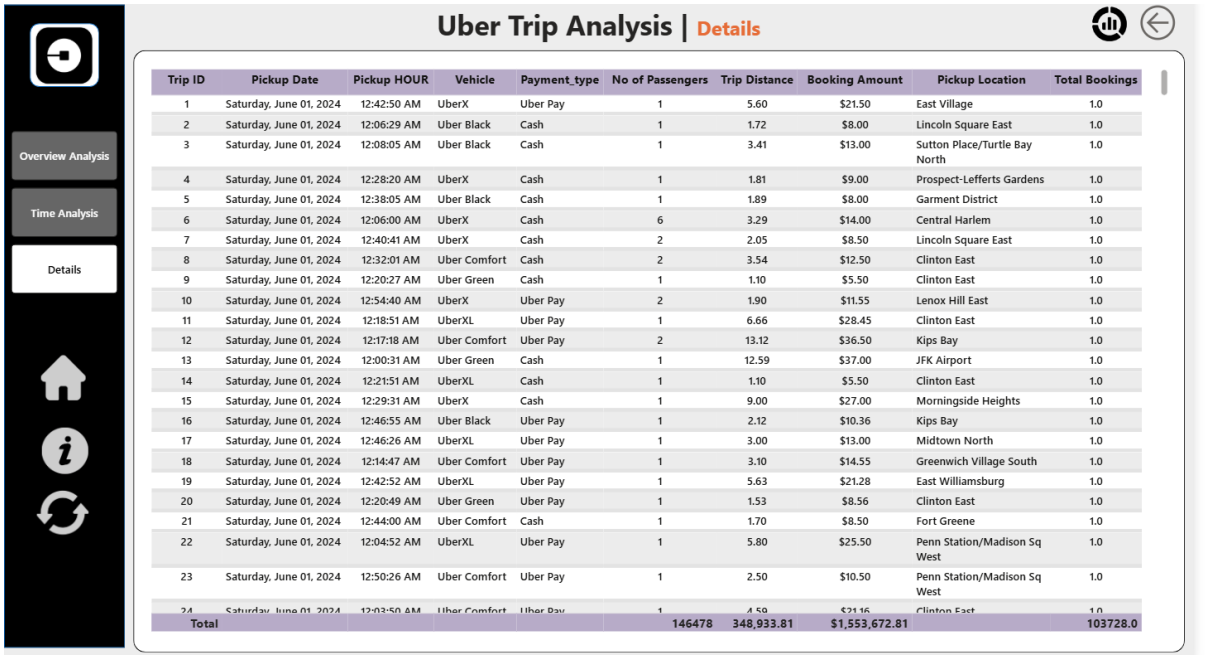
Dashboard 1: Overview Analysis



Dashboard 2: Time Analysis



Dashboard 3: Details Info



Trip ID	Pickup Date	Pickup HOUR	Vehicle	Payment_type	No of Passengers	Trip Distance	Booking Amount	Pickup Location	Total Bookings
1	Saturday, June 01, 2024	12:42:50 AM	UberX	Uber Pay	1	5.60	$21.50	East Village	1.0
2	Saturday, June 01, 2024	12:06:29 AM	Uber Black	Cash	1	1.72	$8.00	Lincoln Square East	1.0
3	Saturday, June 01, 2024	12:08:05 AM	Uber Black	Cash	1	3.41	$13.00	Sutton Place/Turtle Bay North	1.0
4	Saturday, June 01, 2024	12:28:20 AM	UberX	Cash	1	1.81	$9.00	Prospect-Lefferts Gardens	1.0
5	Saturday, June 01, 2024	12:38:05 AM	Uber Black	Cash	1	1.89	$8.00	Garment District	1.0
6	Saturday, June 01, 2024	12:06:00 AM	UberX	Cash	6	3.29	$14.00	Central Harlem	1.0
7	Saturday, June 01, 2024	12:40:41 AM	UberX	Cash	2	2.05	$8.50	Lincoln Square East	1.0
8	Saturday, June 01, 2024	12:32:01 AM	Uber Comfort	Cash	2	3.54	$12.50	Clinton East	1.0
9	Saturday, June 01, 2024	12:20:27 AM	Uber Green	Cash	1	1.10	$5.50	Clinton East	1.0
10	Saturday, June 01, 2024	12:54:40 AM	UberX	Uber Pay	2	1.90	$11.55	Lenox Hill East	1.0
11	Saturday, June 01, 2024	12:18:51 AM	UberXL	Uber Pay	1	6.66	$28.45	Clinton East	1.0
12	Saturday, June 01, 2024	12:17:18 AM	Uber Comfort	Uber Pay	2	13.12	$36.50	Kips Bay	1.0
13	Saturday, June 01, 2024	12:00:31 AM	Uber Green	Cash	1	12.59	$37.00	JFK Airport	1.0
14	Saturday, June 01, 2024	12:21:51 AM	UberXL	Cash	1	1.10	$5.50	Clinton East	1.0
15	Saturday, June 01, 2024	12:29:31 AM	UberX	Cash	1	9.00	$27.00	Morningside Heights	1.0
16	Saturday, June 01, 2024	12:46:55 AM	Uber Black	Uber Pay	1	2.12	$10.36	Kips Bay	1.0
17	Saturday, June 01, 2024	12:46:26 AM	UberXL	Uber Pay	1	3.00	$13.00	Midtown North	1.0
18	Saturday, June 01, 2024	12:14:47 AM	Uber Comfort	Uber Pay	1	3.10	$14.55	Greenwich Village South	1.0
19	Saturday, June 01, 2024	12:42:52 AM	UberXL	Uber Pay	1	5.63	$21.28	East Williamsburg	1.0
20	Saturday, June 01, 2024	12:20:49 AM	Uber Green	Uber Pay	1	1.53	$8.56	Clinton East	1.0
21	Saturday, June 01, 2024	12:44:00 AM	Uber Comfort	Cash	1	1.70	$8.50	Fort Greene	1.0
22	Saturday, June 01, 2024	12:04:52 AM	UberXL	Uber Pay	1	5.80	$25.50	Penn Station/Madison Sq West	1.0
23	Saturday, June 01, 2024	12:50:26 AM	Uber Comfort	Uber Pay	1	2.50	$10.50	Penn Station/Madison Sq West	1.0
24	Saturday, June 01, 2024	12:03:50 AM	Uber Comfort	Uber Pay	1	4.59	$21.16	Clinton East	1.0
Total					146478	348,933.81	$1,553,672.81		103728.0

8. Key Insights :

1. Peak demand occurs during evening hours (commute time).
2. Certain zones show consistently high trip volumes.
3. Cancellation rates increase during high-demand time slots.
4. Revenue is directly correlated with peak-hour demand.
5. A small percentage of drivers handle a large portion of trips.

9. Challenges Faced :

- Handling inconsistent datetime formats.
- Data types (e.g., numeric stored as text) had to be corrected in MySQL.
- Managing large dataset performance in Power BI.
- Data validation across MySQL, Python, and Power BI required careful debugging.
- Mapping location IDs with zone names correctly.
- Ensuring correct supply-demand interpretation.

10. Real-Life Applications :

This dashboard can be used by:

- Operations Team – Optimize driver allocation.
- Strategy Team – Plan surge pricing
- Marketing Teams -- Target high-demand zones.
- Management – Monitor revenue and service quality.
- Data Analysts – Build predictive demand models.

11. Conclusion :

This project demonstrates how transportation data can be transformed into meaningful business insights. By leveraging SQL, data preprocessing, and Power BI visualization, the Uber Trip Analysis dashboard helps in understanding demand trends, improving operational efficiency, and supporting strategic decision-making. It bridges the gap between raw trip data and actional insights.

12. Future Scope :

1. Demand Forecasting Using Machine Learning:

Currently, the dashboard analyzes historical data. In the future, predictive models such as Linear Regression, Random Forest, or LSTM (Time Series Models) can be implemented to forecast:

Future trip demand by hour/day

Expected peak demand zones

Revenue projections

2. Real-Time Data Integration:

The present system works on static CSV data. Integrating real-time APIs or connecting to live databases (SQL Server, Cloud Storage) would allow:

Live trip tracking

Real-time KPI monitoring

Instant operational alerts

3. Driver Performance & Efficiency Score:

A driver efficiency model can be developed based on:

Completion rate

Cancellation rate

Average trip duration

4. Surge Pricing Optimization Model:

Automatically adjust fares during peak hours

5. Multi-Year Comparative Analysis:

Compare performance across multiple cities.

Analyze seasonal trends.

6. AI-Based Recommendation System:

Optimal driver positioning zones

Best pricing strategies

Efficient route allocation

7. Mobile Optimization:

Deploy dashboard on Power BI mobile for anytime access.

13. References:

Kaggle

Uber Dataset

Microsoft Power BI Documentation